



Examining the efficacy of generative artificial intelligence in item generation: comparative analysis of human-developed and AI-generated reading tests

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Abstract

This study examined the utility of questions created with artificial intelligence (AI) compared to those created by human item writers for English reading tests. Despite the rapid adoption of generative AI, such as ChatGPT, in educational settings, there is limited amount of empirical evidence comparing AI-generated and human-generated items based on actual test administration. Given the practical need for empirical validation of AI-generated test items, the present study analysed item difficulty and perceived quality using test data from high school students in South Korea. The findings revealed that AI-generated items were slightly more difficult due to occasional inappropriate language choices, although the difference was not statistically significant. Test-takers generally perceived AI-generated items as comparable to human-developed ones, indicating cautious optimism regarding the integration of generative AI into language test development and preparation.

Keywords ChatGPT · Automated item generation · Item difficulty · Test equation

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1 Introduction

In recent years, AI systems have been proposed as a means to support teaching and learning due to their ability to operate with varying levels of autonomy (OECD, 2019). These capabilities include "detecting patterns in data" and "automating decisions" in various educational processes (U.S. Department of Education, Office of Educational Technology, 2023, p. 1). Among the many AI technologies available, generative AI has gained significant attention, particularly following the introduction of OpenAI's ChatGPT in late 2022. Since its public release, ChatGPT has reached one million users in just five days and two million users within two weeks. This rapid adoption has fueled substantial academic interest, bringing generative AI to the forefront of global discourse (Kim, 2023). Unlike other AI tools that require coding expertise, ChatGPT's ability to generate content effortlessly has made it accessible to a wider audience (see Appendix 1).

Although automated item generation is not a new concept, advancements in AI technology have significantly streamlined the process of creating text, questions, and answers for various types of assessments. According to Yoon et al. (2017), item generation is one of the most essential stages of test development. High-quality, well-designed test items ensure the validity, reliability, and fairness of the assessment. Yoon et al. (2017) highlight the increasing demand for a large pool of diverse and authentic test items, which is a key driver for innovation in item generation practices. Recently, Shin et al. (2025) explored two AI-powered automated item generators—KR-Item-Generator (based on GPT-3.5) and Q-Craft (based on GPT-4)—for creating L2 English reading test items, involving 83 South Korean pre-service teachers. Both tools were found to generate natural and appropriate content, demonstrating strong potential for L2 assessment use. Park and Choi (2025) also suggested potential directions for automated item generation technology based on a study conducted on Korean reading tests. It should be noted that, to date, no study has conducted an empirical experiment involving actual test administration. More empirical evidence is needed to identify the benefits and weaknesses associated with automated item generation.

Building on this, automated item generation technology offers several advantages (Pugh et al., 2016; Shin & Lee, 2023). First, it can drastically reduce the cost of creating assessment items while enabling the rapid development of a large number of test questions. Additionally, it reduces reliance on large teams of test developers, thereby making the process more efficient. Moreover, AI can improve test item management by identifying and removing duplicate questions, ensuring greater variety and quality in assessment materials.

One of the most prominent high-stakes examinations in South Korea, where this study is based, is the College Scholastic Ability Test (CSAT, hereafter). According to a news report, the cost of developing a single CSAT item exceeds 10 million won (\$6,880), excluding administrative expenses (Jeon, 2020). This estimate is based on the daily labor cost of test item developers, but the actual expense is likely to be much higher when additional factors—such as accommodation, food, security, and printing—are considered. While not all assessments require such a significant investment,

the cost of traditional, human-developed test items remains a major financial burden for many testing institutions.

Compared to the extensive research on automated scoring (e.g., Carr & Xi, 2010; Hoang & Kunnan, 2016; Zhang, 2020), relatively few studies have explored automated test item generation since Bormuth's first attempt in 1969. Most of these efforts predate the advent of generative AI and were conducted in the early twenty-first century. Existing research on automated test development has primarily focused on task design, feedback generation, test delivery, and score reporting (Jin & Fan, 2023). However, to date, no studies have specifically investigated the application of generative AI in this domain. While recent research has examined the integration of generative AI in language teaching and learning (Monib et al., 2025; Wu & Ho, 2025), its potential in language assessment remains largely unexplored (Attali et al., 2022).

Addressing this research gap, the present study aims to investigate the potential of generative AI in test item development for language assessment. By analyzing empirical data from real-world test administrations that include items developed by both AI and human test writers, this study seeks to assess the feasibility of using AI-generated content in test construction. With the advent of generative AI, it is now possible to generate complete test items, including stimuli and answer options, with a high level of quality. Ultimately, this research aims to examine the extent to which AI-generated content can be integrated into test development, the processes involved in generating and reviewing items, and the respective roles of human test developers and generative AI (Xi, 2023).

2 Literature review

2.1 Application of AI technology in language assessment

Although still in its early stages, AI technology has been applied to language learning and assessment to explore its effects. Earlier, Xi (2023) stated that enhancing AI literacy among practitioners and users by showcasing recent advancements in AI applications in language assessment is needed. In this regard, Jin and Fan (2023) proposed a model of test-taker engagement in AI-mediated language assessment, which illustrates that test-takers engage with 'task design,' 'assessment delivery,' 'automated scoring,' 'score reporting,' and 'decisions on assessment uses' as essential phases of developing and validating language tests (p. 492). Within this model, AI-based item development should also be taken into account, in combination with task design.

Until recently, studies involving AI technology in language assessment have mainly focused on automated writing evaluation (AWE). In a recent special issue on AWE, Ranalli and Hegelheimer (2022) stated that there has been ongoing interest in how automated evaluation technology can support second language (L2) learners, teachers, and practitioners in the field of computer-assisted language learning (CALL). A recent study by Wang et al. (2024) investigated the impact of iWrite, an AWE program, on autonomous learning among 146 non-English major undergraduates at a public university in China. Using a convergent parallel mixed-methods

approach, the study explored students' perceptions of AWE in self-directed learning environments. While students responded positively to language-based feedback, their perceptions of content-based feedback were less favorable. Another study on iWrite found that undergraduates in China were generally satisfied with its use as a surrogate writing tutor, demonstrating increased engagement and self-regulation over a 16-week period (Wang, 2025). Both studies, however, highlight areas for improvement in the design of AWE system.

Regarding the use of AI in speaking assessment, recent research examined the accuracy of automated speech recognition (ASR) systems and showed that they are significantly affected by the type of oral tasks and the language backgrounds of students (Hannah et. al., 2022). Studies that employ chatbot in speaking assessment include using spoken dialog system (SDS) to investigate test-taker's preference (Chukharev-Hudilainen & Ockey, 2021), developing speaking tasks to converse with AI-based voice chatbots (Kim & Lee, 2023), and practicing controlled language drills with L2 elementary-level students (Yang et al., 2019).

While studies have been conducted actively on using AI in evaluating productive skills and test-takers' engagement with tasks, little has been researched about utilization of AI in test development. Lee et al. (2023) introduced AI-generated content using CopyAI to enhance the reading experience for young English learners. In their study, a total of 121 participants were divided into AI text and traditional reading groups and their reading comprehension ability were compared after the experiment. Significant improvements were found in the AI group, with around 80% finding it more engaging and over 75% reporting enhanced concentration with AI texts. Lee et al. (2023) suggest that AI-generated texts can be provided as extensive reading resources for language learners. On the other hand, Hellmich and Vinall (2023) highlighted the need for more longitudinal studies to determine the ongoing benefits of AI-powered writing assistants.

The rapid rise of ChatGPT and AI applications in language assessment has prompted a need for new theories and best practices. Also, the new generative AI technology has made it possible not only to recognize and evaluate answers but also to generate test items. Therefore, more studies are required to explore the potential of using AI technology in developing test items, as well as to validate and redefine language test constructs to reflect real-world communication changes (Xi, 2023).

2.2 Studies on automated item generation (AIG)

Research to date has shown that AI performs at a level comparable to humans in generating text and answering test items, demonstrating significant logical competence (Lee et al., 2023). However, there are many other factors that affect the development of a test item, such as the attractiveness of the answer choices, which plays a crucial role in determining the difficulty of the test item. In Shin and Lee's (2023) study, human and AI-generated texts were not statistically different in terms of the naturalness of English expression and story flow, but they were significantly different in terms of the attractiveness of incorrect answers, indicating that AIs such as ChatGPT are not yet as good as human test writers. Nevertheless, the above comparisons of AI's and humans' performance in developing test items were based on perception

surveys of learners or experts, and little research has used empirical data from actual tests.

The introduction of ChatGPT has made the automated item generation technology more accessible to the general public. OpenAI's GPT-3 model, aligned with the transformer paradigm, excels at generating new and coherent content based on minimal prompt examples. In particular, GPT-3's capabilities obviate the need for explicit model updates such as annotation tagging, model development, or iterative training processes (Brown et al., 2020). Impressively, GPT-3 can mimic numbered lists, adapt to different input text styles, formulate comprehension queries, compose symmetrically structured paragraphs, and even produce structured HTML code, leading to outputs comparable to those produced by skilled human writers. In the same vein, Shin and Lee (2023) utilized the item types found on the College Scholastic Ability Test, the South Korean college entrance exam, to create English test items. Using ChatGPT 3.5, only prompts and a model test item were entered to closely mimic the actual test items. Shin and Lee (2023) reported that automated item generation through generative AI is possible with no-code applications. However, there are also vulnerabilities in using such generative AI for test development. Bolender et al. (2023) pointed that attempts to develop tests using generative AI are often carried out by "startups and tech gurus" for their monetary gains, without conducting sufficient research to account for test takers' knowledge, skills, and abilities in a valid manner (p. 513). They suggested that careful integration of generative AI by testing agencies is required until AI performance improves sufficiently.

Attali et al. (2022) explored transformer-based Automatic Item Generation (AIG) for reading tests, addressing the limitations of traditional rule-based models. Leveraging GPT-3, the Interactive Reading (IR) task generates reading passages, comprehension questions, and automated scoring, enhancing efficiency in educational assessments. The study conducted a large-scale pilot with 200,000 test sessions and confirmed high validity, efficient item discrimination, and reduced response times due to the interactive format. However, Attali et al. (2022) claimed that an important limitation of the item model approach and its conventional implementation is its restriction to content areas that are relatively easy to model, such as math. In addition, they noted that because the process relies on highly skilled content experts to create models, AIG remains semi-automatic and resource-intensive, making large-scale deployment costly.

This study aims to examine how useful AI tools can be in the development of test items and to what extent they can guarantee the quality of test items compared to those developed by humans. In other words, the objective of this study is to examine whether generative AI technology might produce items with similar quality and difficulty to those previously created by human writers. Among several factors associated with item difficulty in English reading tests, the present study mainly focuses on two key factors identified in previous research: text complexity and task complexity (Freedle & Kostin, 1993).

This present study formulates the following research questions:

RQ1. To what extent does the difficulty level of AI-generated English reading test items differ from that of human-developed test items?

- RQ2. How do task type (e.g., literal comprehension, inferencing) and item type influence the difference in difficulty between AI-generated and human-developed test items?
- RQ3. How do participants perceive the difficulty of AI-generated items compared to human-developed items? Do participants consider AI-generated items to be guaranteed of comparable quality and difficulty?

3 Method

3.1 Participant

A total of 268 students from 10 classes of four public high schools of South Korea participated in this study. In high schools of South Korea, where the allocation of students by class is structured to ensure equivalent academic levels across each class, a two-group division has been implemented, as shown in Table 1. This division is based on the class unit, with Group A receiving a test form developed by human item developers and Group B receiving a test form generated by AI.

In addition, all participants were in the 11th grade with similar English learning experience (officially 9 years starting in the 3rd year). All participants had been studying English as a foreign language, but none had an experience of living or studying abroad when the study was conducted. The participants' English proficiency levels were generally B2 to C1 levels under CEFR framework (Lee & Shin, 2019). As potential test-takers of the CSAT, most participants were familiar with the types and difficulty level of the test items when the experiment was conducted.

3.2 Instrument

In this study, two test sheets were developed and tested for the purpose of comparing the characteristics of test items created by human test item writers and test items created by AI. The A-type test sheet consists of items created by human test item writers, while the B-type test sheet consists of items developed by the researchers using ChatGPT. In this study, we use 'reading items' as an umbrella term for both the reading passage and the question-and-answer options. For this, 28 English reading items (No. #18 to #45) from the 2020 CSAT were used as A-type test sheet. Currently, all CSAT items are written and reviewed by human developers, consisting of university professors of English-related fields and high school English teachers in Korea. Most of them are highly trained item writers who had sufficient experience of developing English test items. They lodge at an undisclosed place to develop the test items for approximately 2–5 weeks. During this time, extremely rigorous item writ-

Table 1 Organizing experiment groups

Group	Group A (School-Class): $n = 141$	Group B (School-Class): $n = 127$
School- Class	A-1, A-2, B-3, C-1, D-1	A-3, B-1, B-2, C-2, D-2

^aAutomated Item Generation

^aText-to-Image tools

ing and review processes are conducted to prevent any potential errors in the items. The CSAT English test has a total of 45 test items, 17 listening and 28 reading items (No. #18 to #45). This study focused on the development of reading items. One of the 28 items was a question asking for content mismatching between a graph and its descriptive text, but since ChatGPT did not have the ability to accurately produce a graph based on the description at the time of the experiment, it was excluded from the study. In addition, to compare the difficulty level of the original CSAT items produced by human experts and the ChatGPT-generated items, we applied a test equating technique to equalize the two test forms. To this end, five anchor items in each test form were used in common. Therefore, the last five items of the original test form were added to the test form composed of ChatGPT-generated items. Thus, the total number of test items generated by ChatGPT, B-type test sheet, was 22 (No. #1 to #22).

The process of developing test items using ChatGPT followed the manual proposed by Shin (2023), who introduced a total of nine prompts to develop reading items for the CSAT English test. The test item development process proposed by Shin (2023) has been validated by follow-up studies (e.g., Kim, 2024; Park & Choi, 2025). According to Shin (2023), simple prompts are used to generate test items in the prompts such as "To find the purpose/gist/intent/topic/title of the passage" or "To find the agreement/disagreement of the details," where the format of the options does not have a significant impact on the item type. This is because ChatGPT, which uses large language model, demonstrates an outstanding performance in summarizing texts into concise phrases or sentences. Therefore, relatively simple prompts were used when these common items as shown in Table 2 below.

On the other hand, if the type of the test item, especially the organization of the choices, is closely tied to the characteristics of the passage, then item-specific prompts are needed. For such complex tasks, more prompts are required to generate sophisticated conditions. Table 3 below contains the successful prompts for a variety of item types.

As presented in Table 3, a total of eight different prompts were used to generate passages for each item type. For example, on items that asks test-takers to find the order of the text, specific commands were used such as "keep the first sentence... shuffle paragraphs...". The following Table 4 shows an example of creating a new test item directly with ChatGPT for the type of "To find a sentence that is inappropriate for the flow of the text."

As shown in Table 4, generating a complex task requires a substantial length of prompts. In addition, a model passage needs to be provided to ChatGPT to generate a similar level text. For effective test equating, 5 items were used as anchor items, which are approximately 20% of the total items. Generally, anchor items equating

Table 2 Prompt for common test items (Shin, 2023, p. 31)

Create a new passage with a different topic and make a MCQ (including 5 choices)

1) to test the understanding of the purpose/gist/main idea/best title of the passage

2) to confirm the agreement with the details of the passage as follows: Add a model test item here

※ For "To find agreement/disagreement with the details," the user can optionally specify additional conditions, such as "Only one answer choice is true/false."

Table 3 Item-specific prompts (Shin, 2023, p. 31–32)

Type of test item	Prompt sequence
1. To identify the mood and atmosphere	Create a new passage with a different topic and make a new MCQ as follows: Add 'question,' 'passage,' and 'choices' of the target type of the test item here
2. To understand the contextual meaning of the underlined expression	1) Create a new passage with a different topic as follows: Add 'passage' of the target type of the test item here 2) Underline one phrase and create five choices for what that phrase implies in the following passage: Add a passage generated by AI here
3. To infer an appropriate phrase for a blank	Create a new passage with a different topic as follows and make a <> by deleting one phrase, and make 5 choices for <>: Add 'passage' of the target type of the test item here (Replace an underline with <> or)
4. To find a sentence that is inappropriate for the flow of the text	Create a new passage with a different topic, and keeping the first two sentences of the passage, then assign 5 numbers before some sentences that make up the rest part of the passage, additionally only one of the 5 sentences should be unsuitable for the flow of the text as follows: Add 'passage' of the target type of the test item here
5. To order the text	1) Create a new passage with a different topic as follows: Add 'passage' of the target type of the test item here 2) Keep the first sentence in the following passage and shuffle paragraphs (A), (B), (C): Add a passage generated by AI here 3) Revise the above passage by adjusting the amounts in paragraphs (A), (B), and (C) to be equal. (If necessary)
6. To insert a sentence to fit the flow of the text	Paragraph (A) is taken from one of (1), (2), (3), (4), (5) in the following passage, and make a new passage with a different topic as the following passage and extract one paragraph like (A) at the top of the story and make 5 (s) as options: Add 'passage' of the target type of the test item here
7. To find a grammatical error	1) Create a new passage with a new topic as the following passage and make five <>s containing a word: Add 'passage' of the target type of the test item here 2) Change one of the words in the five <> into an unsuitable or ungrammatical word out of context of the following passage: Add a passage generated by AI here
8. To infer appropriate words to fit the blanks in a passage summary	1) Create a new passage with a different topic as follows: Add 'passage' of the target type of the test item here 2) Summarize the following passage in one sentence or so: (<i>Insert the passage</i>) 3) Make it shorter. (If necessary) 4) Make blanks (A) and (B) for two key words in the summary 5) Make 5-word pairs as options for (A) and (B)

MCQ stands for 'multiple choice questions'

20% or above of the total items are acceptable (De Ayala, et al., 2018; Kolen & Brennan, 2004). For more specific information on item-generation procedure, see Appendix 2.

An online questionnaire consisted of 7 questions, developed by the first author and reviewed by co-authors, was used. After a series of reviews and revisions, we created a final draft of the questionnaire using Google Forms and used it to investigate the perceptions of the participants who took the B-type test sheet to further investigate their perceptions about the items they received (see Appendix 3).

3.3 Data collection

After constructing a B-type test sheet with the questions generated by ChatGPT, both test sheets were given to the participants to examine the difficulty levels of the items.

Table 4 Sample prompt to create finding the appropriate sentence item

Prompt Including a Model	다음 글에서 전체 흐름과 관계 <u>없는</u> 문장은? (Which sentence in the following passage is inappropriate for the flow of the text?)
Test Item	Answer: ㉓ Create a new passage with a different topic, and keeping the first two sentences of the passage, then assign 5 numbers before some sentences that make up the rest part of the passage, additionally only one of the 5 sentences should be suitable for the flow of the text as follows: Although commonsense knowledge may have merit, it also has weaknesses, not the least of which is that it often contradicts itself. For example, we hear that people who are similar will like one another (“Birds of a feather flock together”) but also that persons who are dissimilar will like each other (“Opposite attract”). ㉑ We are told that groups are wiser and smarter than individuals (“Two heads are better than one”) but also that group work inevitably produces poor results (“Too many cooks spoil the broth”). ㉒ Each of these contradictory statements may hold true under particular conditions, but without a clear statement of when they apply and when they do not, aphorisms provide little insight into relations among people. ㉓ That is why we heavily depend on aphorisms whenever we face difficulties and challenges in the long journey of our lives. ㉔ They provide even less guidance in situations where we must make decisions. ㉕ For example, when facing a choice that entails risk, which guideline should we use—“Nothing ventured, nothing gained” or “Better safe than sorry”?\$
Generated Test Item	다음 글에서 전체 흐름과 관계 <u>없는</u> 문장은? (Which sentence in the following passage is inappropriate for the flow of the text?) Answer: ㉓ Although traditional beliefs may have some validity, they also have limitations, not the least of which is that they often contradict each other. For example, we hear that honesty is the best policy (“Honesty is the best policy”) but also that it is sometimes better to be discreet (“Discretion is the better part of valor”). ㉑ We are told that hard works pays off (“No pain, no gain”) but also that working too hard is detrimental to one’s health (“All work and no play makes Jack a dull boy”). ㉒ Each of these conflicting beliefs may be true I certain circumstances, but without a clear understanding of when they apply and when they don’t, they offer little guidance in navigating life’s challenges. ㉓ That is why we often rely on these beliefs in moments of uncertainty. ㉔ However, they are even less useful in situations where we must make important decisions. ㉕ For example, when faced with a difficult choice, how do we decide between “The early bird catches the worm” and “Good things comes to those who wait”?

Among the entire sample of 268, 141 were given the type A test sheet, and the rest 127 of them were given the type B test sheet. The allocation of the test type was carried out randomly, assigning one of the types of test sheet to intact classes. Each test was administered for 45 to 50 minutes, the entire duration of an English lesson at a high school. The participants who were given the type B test sheet were asked to respond to a questionnaire immediately after the test finished. The participants’ answers to both the test and the questionnaire were computed for data analysis. The following Fig. 1 illustrates this data collection procedure.

3.4 Data analysis

The analysis of data centered on the interrelationship of items, encompassing both types of test sheets. For this, the present study used the common-item non-equivalent groups design. In this design, item difficulties can be calibrated on the same scale using the difficulties of five common items. The difficulty level of each test sheet, as a whole and by task types and item types, was computed through the Rasch model

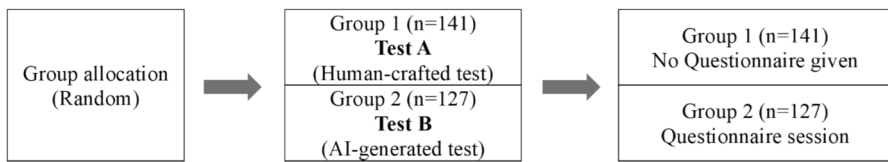


Fig. 1 Data collection procedure. Note.
Source: Author's own work

Table 5 Descriptive statistics of item difficulty indices

	Total	Test A(Human)	Test B(ChatGPT)
Mean	−0.16	−0.42	−0.02
SD	1.20	1.11	1.30
Max	2.09	0.94	2.09
Min	−3.20	−3.20	−3.04

SD indicates standard deviation index

based concurrent calibration using the IRTQ package in R (version 4.2.3). Specifically, we used the Rasch model for estimating item difficulties. Since we had two test forms and five anchor items in each test form, we used concurrent calibration for equating, where item difficulties in Test A and B were estimated simultaneously, and anchor items were constrained to be estimated equal across Tests A and B. By comparing the mean values of the item difficulty measures of the two test sheets, we aimed to investigate the extent to which the AI-generated English reading items might be more or less difficult than the traditional human-developed item. To examine the item difficulty differences between the two test forms, we conducted independent t-test using the software Jamovi (The Jamovi Project, 2023). In addition, we compared the text complexity of the two test forms using ETS's TextEvaluator® (ETS, 2010). We analysed the text of all 22 items of each test form, excluding the five common anchor items. This text analysis enabled us to identify the most challenging source of comprehension difficulty out of eight possible sources. Ultimately, we aim to explore the potentials of utilizing the generative AI tool to develop English reading items.

4 Results

4.1 Analysis on item difficulty

The findings in Table 5 show that the A-type test set, featuring questions crafted by human item writers, were easier than the B-type test set comprising items generated by the ChatGPT. The mean item difficulty measure of the A-type test was −0.42, which is slightly lower than the mean difficulty measure of B-type test, −0.02. Also, the maximum item difficulty measure of B-type test was 2.09, which is substantially larger than the maximum measure of A-type test, 0.94. When independent t-test was conducted to compare the mean item difficulty levels between two test sets, it was found that the difference was not statistically significant: $t(42) = -1.05$, $p = 0.30$. Therefore, it can be seen that a few items have very difficult item difficulty measures

in B-type test, which is crafted by ChatGPT, but the difficulty level as a whole was not statistically significant (Fig. 2).

4.2 Analysis on text difficulty

To examine the difference found in text difficulty of the two test forms, we compared the TextEvaluator® Complexity Score (TECS) of the passages. It provides an overall measure of text complexity by analyzing Syntactic Complexity (SC), Academic Vocabulary (AV), Word Unfamiliarity (WU), Concreteness (C), Lexical Cohesion (LC), Interaction/Dialogue Style (ICS), Level of Argumentation (LA), and Degree of Descriptiveness (DN) as sources of comprehension difficulty.

Some text complexity variables showed significant correlations with item difficulty when analyzed independently. Among the listed text complexity variables, overall text complexity score (TECS), degree of specificity as a descriptive style (C) (Coltheart, 1981), and academic vocabulary frequency (AV) (Coxhead, 2000) showed significant relationship with item difficulty. Pearson's correlation coefficients of TECS, C, and AV with item difficulty were $r=0.602$ ($p<0.001$), $r=-0.439$ ($p<0.003$), and $r=0.525$ ($p<0.001$). However, no significant difference was found in these text complexity variables between the two test forms. The overall text complexity score (TECS), for example, averaged 1080.86 for A-type test and 1095.41 for B-type test, indicating that the average passage difficulty for both test forms was in the K-10 level based on TextEvaluator to Common Core Concordance (Sheehan, 2015). Although the descriptive statistics suggest that ChatGPT's passages may be slightly more difficult, the difference was not statistically significant (see Table 6).

As aforementioned, there were no statistically significant differences in the text complexity scores between the two test forms. This finding demonstrates that ChatGPT can generate items by mimicking the item developed by a human item developer, while maintaining a level of difficulty that is quite similar to that of a human item developer.

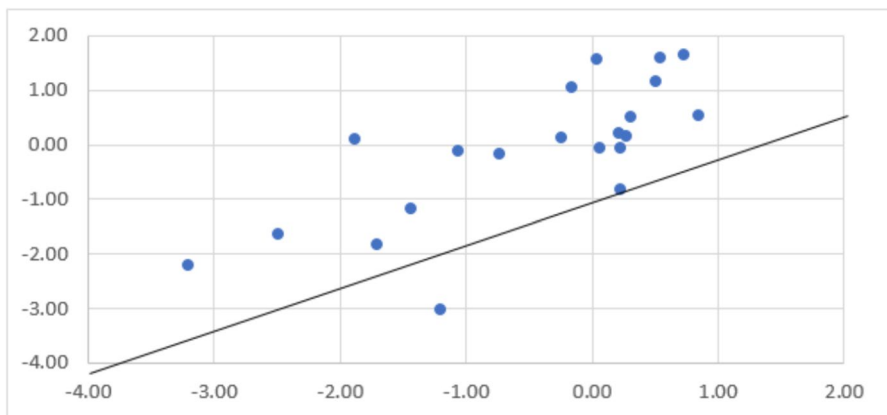


Fig. 2 Item difficulty level correlations between the test sets. Note. Source: Author's own work

Table 6 Descriptive statistics of text complexity

		SC	AV	WU	C	LC	ICS	LA	DN	TECS
Test A	M	61.23	72.91	77.45	33.05	44.82	29.59	45.91	39.59	1080.86
(Human)	(SD)	(13.70)	(16.19)	(12.12)	(15.63)	(9.21)	(20.48)	(21.12)	(21.43)	(269.94)
Test B	M	61.55	78.64	75.05	30.64	46.5	27.09	33.77	31.36	1095.41
(GenAI)	(SD)	(13.57)	(13.61)	(10.25)	(13.6)	(9.09)	(21.26)	(25.29)	(24.81)	(222.57)
t-score		0.08	1.24	-0.70	-0.53	0.60	-0.39	-1.69	-1.15	0.19

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 7 Item difficulty indices by task types

Test	Task type	No. of items	Mean	Min	Max
Total	Literal comprehension	6	-1.24	-1.87	0.08
	Inferencing	38	-0.04	-3.20	2.09
Test A(Human)	Literal comprehension	3	-1.67	-1.87	-1.44
	Inferencing	19	-0.22	-3.20	0.94
Test B(ChatGPT)	Literal comprehension	3	-0.98	-1.82	0.08
	Inferencing	19	0.13	-3.04	2.09

According to the scatter plot presented in Fig. 1, the correlation coefficient between the difficulty levels of questions in the A-type and B-type tests stood at approximately 0.768. This relationship underscores certain distinctions in the overall difficulty across the test sets. However, it is important to note that despite these differences, the hierarchical order of question difficulty remained notably consistent between the two test types. It should be noted that anchor items were not included in the correlation analysis.

4.3 Analysis on item difficulty by task type

To investigate whether item difficulty might vary by task type (literal comprehension and inferencing), mean item difficulty indices were computed separately for each task type within the test as a whole, and for each individual test type (See Table 7).

The findings in Table 7 show that the item difficulty of each task type was generally easier with literal comprehension items, showing a lower difficulty level index (-1.24) than inferential understanding items (-0.04). Similar results were also found in the analysis of each individual test sheet. For example, on the A-type test sheet, literal comprehension items were found to be easier (-1.67) than inferential understanding items (-0.22). On the B-type test sheet, literal comprehension items were also found to be easier (-0.98) than inferencing items (0.13). Hence, there was no difference in item difficulty according to the type of task between the traditional human developed test (Test A) and the ChatGPT generated test (Test B) (Fig. 3).

4.4 Analysis on item difficulty by specific item type

To further investigate whether item difficulty might vary by eleven item types, mean item difficulty indices were computed for each item type separately. The results presented in Table 8 below indicate that 'identifying the flow of the text item' type

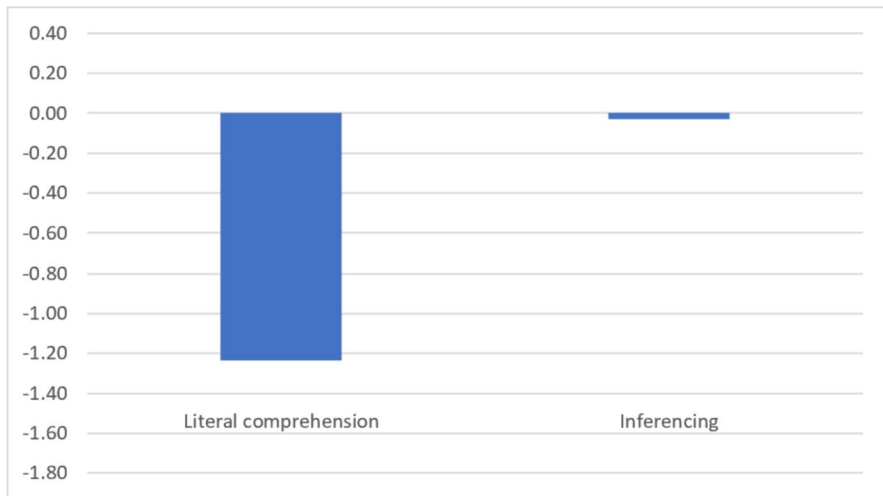


Fig. 3 Item difficulty by task type. Note.

Source: Author's own work

Table 8 Item difficulty indices by each item type

Specific item type	Total	Test A (Human)	Test B (ChatGPT)
Purpose of the text	-2.71	-3.20	-2.22
Order of the text	0.24	0.53	0.24
Summary of the text	-0.45	-0.73	-0.17
Title of the text	0.45	-0.15	1.05
Topic of the text	-1.21	0.22	-1.93
Flow of the text	0.77	0.35	1.19
Word usage within a context	0.31	0.22	0.20
Word meaning	-0.59	-1.06	-0.12
Fill-in-the-blank (Cloze)	0.72	0.47	0.97
Changes in characters' feeling	-2.07	-2.49	-1.65
Grammar	-0.06	-0.24	0.12

showed the highest difficulty index (0.77), while 'finding the 'purpose of text' type was the easiest, with the lowest difficulty index (-2.71). Other item types that showed relatively high difficulty indices are: 'order of the text', 'title of the text', and 'fill-in-the blank.' Item types that showed comparably low difficulty indices were: 'topic of the text' 'word meaning', and 'changes in characters' feeling.' The following Table 8 and Fig. 4 present the results of the difficulty analysis for each item type.

We also looked at whether different trends might be found between the two types of test sheets, but the findings showed that they were generally very similar (see Fig. 5). An exception was found in one of the item types, finding the topic of the text items, where a slight difference in item difficulty was found between the two tests. The finding shows that the topic of the text items appears to be much more difficult in the type A test (0.22) than in the type B test (-1.93).

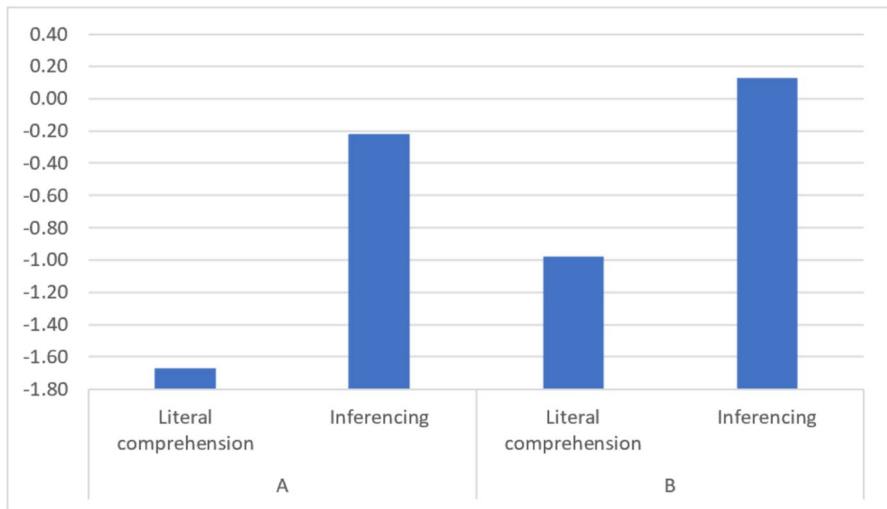


Fig. 4 Item difficulty by task type (each test sheet separately). Note. Source: Author's own work

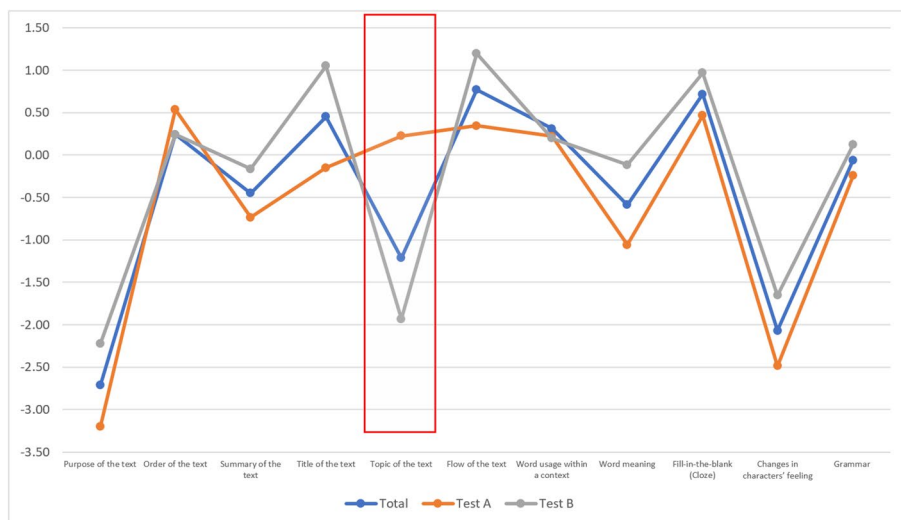


Fig. 5 Item difficulty by item type (each test sheet separately). Note. Source: Author's own work.

4.5 Questionnaire results

To investigate the test-takers' perceptions and beliefs of the AI-generated reading items, we administered a questionnaire to those who took the B-type test. The questionnaire asked them to assess the difficulty level of the items while disclosing that they were AI-generated. The following Table 9 presents the test-takers' perceptions about the AI-generated English reading items reported by the participants.

Table 9 General impressions about the AI-generated items

Q. The items in the test B were all written by an Artificial Intelligence (except the last 5 items). How did you feel about the items?	Frequency	Rate (%)
I thought the items were written by some non-native speakers of English because they were very awkward	4	3.15
I thought the items were not written by teachers because they were somewhat awkward	38	29.92
They were normal	44	34.65
I thought the items were written by teachers because they were generally natural	30	23.62
I thought the items were written by professional item writers because they were very natural	11	8.66
Total	127	100

Table 10 Perceptions about item difficulty level

How did you feel about the difficulty level of the items written by the AI, compared to those CSAT items?	Frequency	Rate (%)
They were very easy	1	0.79
They were somewhat easy	32	25.20
They were similar	47	37.01
They were somewhat difficult	38	29.92
They were very difficult	9	7.09
Total	127	100
(Question for those who answered ‘very easy’ and ‘somewhat easy’)	Frequency	Rate (%)
What type of item were particularly easy?		
Vocabulary	8	18.60
Sentence structure	12	27.91
Content	10	23.26
Item complexity (answer option design)	13	30.23
Other	0	0.00
Total	127	100
(Question for those who answered ‘very difficult’ and ‘somewhat difficult’)	Frequency	Rate (%)
What type of item were particularly difficult?		
Vocabulary	6	10.91
Sentence structure	10	18.18
Content	17	30.91
Item complexity (answer option design)	22	40.00
Other	0	0.00
Total	127	100

The findings show that the highest number of responses was “they were normal” (34.65%), which indicates that one third of the participants perceived the items natural. 29.92% of the participants felt that the items were “somewhat awkward” while 23.62% of the participants felt that the items were as natural as those written by the teachers. 8.66% of the participants reported that the items were “very natural” and “professional.” Only 3.15% of the participants felt that the items were very awkward. We further asked about perceived difficulty levels of the AI-generated items, compared to the traditional CSAT items, and the findings are presented in Table 10 below:

Table 11 Potential advantages of the AI-generated items

If AI can spontaneously generate items similar to the tests you just took, would they be helpful for your test preparation?	Frequency	Rate (%)
Not helpful at all	3	2.36
Not helpful	23	18.11
Neutral	33	25.98
Somewhat helpful	55	43.31
Very helpful	13	10.24
Total	127	100

According to the results presented in Table 10, the largest proportion of the participants (37.01%) responded that they found the difficulty level of the AI-generated items was generally ‘similar’ to the human-developed CSAT items. A total of 37.01% of participants reported the AI-generated items as more difficult (somewhat difficult and very difficult) than the original CSAT items, and 25.20% of participants perceived the items as easier (somewhat easy and very easy) than the CSAT items. We asked a follow-up question about what specific item factor made the items easier for those who answered that the AI-generated items were easier, and 27.91 % answered that items related to sentence structure and 30.23% of them chose complexity of item structure (answer option design). On the other hand, when we asked what item factor made the items particularly difficult to those who answered that the AI-generated items were difficult, 40% of them reported item complexity, and 30.91% of them answered content of the prompt. It should be noted that the participants were all very familiar with the CSAT items as they had been exposed to the mock CSAT and retired items of the CSAT as part of the preparation for the exam.

Lastly, we asked about potential advantages of using the AI-generated items in preparing for the target tests and the results are presented in Table 11 below. Findings show that 43.31% of the participants felt it would be somewhat helpful, and 10.24% of them also responded that it would be very helpful. In contrast, 18.11% found it would be not helpful, and 2.36% of the respondents felt it would not be helpful at all. Substantial numbers of respondents showed a neutral perception on this question (25.98%).

5 Discussion

5.1 Interpretations of findings

In order to explore the potential utility of questions generated by ChatGPT, a comparison was conducted between the questions developed by human item writers and those generated by ChatGPT. This comparison encompassed the mean difficulty level indices of the test sheets, indicating the trends in question-specific difficulty, and the difficulty levels of questions categorized by task type and specific item type. The results corresponding to Research Question 1, *To what extent does the difficulty level of AI-generated English reading test items differ from that of human-developed test items?*, are as follows. It was found that items generated by ChatGPT showed

higher levels of difficulty compared to those developed by human item writers. However, the difficulty measure gap between the two tests was not statistically significant ($p > 0.05$). Therefore, we cannot argue that the items generated by ChatGPT are substantially more difficult than the traditional human-developed items. It should be noted that there are several factors that affect the item difficulty of English reading tests but generating the reading items using ChatGPT is technically a replication of the reading stimuli and answer options. Task type and item type remained the same, and the length and style of writings (i.e. sentence complexity, etc.) were generally similar to the original CSAT stimuli, following the commands and guidelines provided by the researchers. Only uncontrolled factors in developing the reading items through ChatGPT were specific choices of words, and genres (topic) of the stimuli. Common factors affecting item difficulty suggested by a number of studies are lexical complexity, syntactic pattern density, number of words in a sentence, level of inferencing required to answer the item, and readability index (Freedle & Kostine, 1991; Hwang & Lee, 2020; Kwon & Shin, 2022). Therefore, we speculate that the vocabulary used in the AI-generated reading stimuli might have caused the slight increase in item difficulty measure. For example, in the AI-generated reading stimuli, some words such as “discretion,” “valor,” or “detrimental” are outside the National curriculum vocabulary list and beyond the level used in retired CSAT items. Normally, human item writers of CSAT and other relevant English tests use original texts from various sources and revise and edit them to create a reading stimulus. During the procedure, they make careful decisions on language choices, such as sentence length and choice of words to control syntactic and lexical complexity. However, the reading stimuli generated by ChatGPT lack such “human touches” to control the sophisticated elements that affect item difficulty. Hence, the generative AI does not necessarily follow the standards commonly employed by human item writers. We speculate that such difference was the cause of the higher item difficulty measure shown in the B-type test.

The results corresponding to Research Question 2, *How does task type (e.g., literal comprehension, inferencing) and item type influence the difference in difficulty between AI-generated and human-developed test items?*, are as follows. We found that the distinction in overall item difficulty measure was not confined to specific questions. Rather, it was systematically present across the entire test. Such systematic variations in difficulty were also evident in the analysis of task types and specific item types. The findings of the analysis on task type indicated that the inferential items showed higher levels of difficulty than the literal comprehension items. This trend was consistent across both tests. However, we believe that this consistent finding stems from the level of difficulty caused by the type of the task per se, rather than by different item writers (humans vs. AI). In fact, inferencing items are originally more difficult than the literal comprehension, as suggested by several existing studies (Freedle & Kostine, 1991; Jang, 2004). This finding suggests that ChatGPT generates both inferential and literal comprehension items with a similarity closely resembling that of human item writers. In other words, we found that ChatGPT can develop reading items that show generally consistent difficulty levels as human item writers. Furthermore, in the analysis of specific item types, except for the finding the topic of the passage items, the trends in item difficulty across all item types were remarkably

similar between the two tests. Again, this implies that items generated by ChatGPT retain similar level of item difficulty in 10 out of 11 item types. Only an exception was found in one item type, “finding the topic of a text.” The disparity observed in this item type can be attributed to the specific commands utilized to generate new passages with ChatGPT. As aforementioned, the command prompts we used in most items were “*Create a new passage with a different topic.*” Thus, ChatGPT has generated new passages with similar length, difficulty, and style of writing but with a different topic from the original passage we inserted. What algorithms the ChatGPT has used to find and apply to the new passages is unknown, leading us to speculate that the selection of new topics is random and beyond our control.

The results corresponding to Research Question 3, *How do participants perceive the difficulty of AI-generated items compared to human developed items? Do participants consider AI-generated items guarantee to be of comparable quality and difficulty?*, are as follows. The findings show that while a greater number of participants found AI-generated items natural and normal, a few others still found them awkward. The majority believed they were similar in difficulty to human-developed items, with some finding them more challenging. Perceived ease or difficulty was often linked to factors like sentence structure and item complexity. Based on these findings, it is apparent that certain items exhibit a lack of naturalness in terms of sentence structure and item design. In particular, crafting answer options is a highly intricate task for human item writers since it directly impacts item difficulty. The degree to which distractors are attractive is closely related to the difficulty level, and therefore, human item writers of various high-stake standardized tests give significant considerations to this factor. However, generative AI such as ChatGPT at the current level does not seem to consider this factor as carefully as humans. Since the attractiveness of distractors is somewhat a subjective standard, it could be difficult to train the AI to such sophisticated aspects. Also, possible bias and sensitivity issues in AI-generated items may be problematic, so that careful attention is required by test practitioners when utilizing generative AIs in test development (Xi, 2023). Therefore, further human adjustments are essential after generating stimuli using generative AI tools.

When asked about the potential benefits of using AI-generated items for test preparation, 43.31% thought they would be somewhat helpful, while 10.24% found them very helpful. However, 18.11% did not find them helpful, and 2.36% deemed them not helpful at all, with 25.98% having a neutral opinion. From this finding, we can speculate that the majority of participants admit the potentials of using the AI-generated English reading items as practice test purposes. Since the participants were potential test-takers of CSAT, which is an extremely high-stakes exam in Korea, we assume that the idea of replicating new CSAT items can be a good practice for them. Building on earlier findings, which indicated a higher number of participants perceiving the items as natural and normal, the majority of participants viewed the AI-generated items as authentic as human-crafted ones, potentially enhancing their value for exam preparation. Consequently, it can be inferred that these items hold promise for inclusion in actual tests, given that most participants found them genuine and natural.

5.2 Implications

The present study proposes a number of implications and considerations for developing tests using generative AI in the future. Firstly, we found a great potential of utilizing ChatGPT for the development of tests equivalent to conventional tests, showcasing its potential for practical application in the development of standardized assessments. This is particularly demanding in situations where discrepancies in question difficulty across different years have raised societal concerns, as observed in exams like Korea's CSAT. Based on the findings, we found that generating items with a similar level of difficulty through generative AI is feasible at the current stage, and the potential for further improvement in language model generation performance can be achieved as technology advances. We contend that generative AI, like ChatGPT, exhibits sufficient performance to execute the statistical method proposed by Kurdi et al. (2020). ChatGPT's machine learning techniques are grounded in natural language processing, leveraging extensive datasets (Brown et al., 2020). Consequently, this new approach to generative AI-based item development opens a new possibility of automated item development. If generative AIs can consistently demonstrate comparable performance in item development, but in a more efficient and cost-effective manner, they have the potential to replace human item writers in the near future.

Secondly, although item generation performance demonstrated by ChatGPT has great potential, we believe that additional "fine-tuning" job by human is still required at the finalizing stage. As aforementioned, some of the factors, such as choice of words and answer option designs need more careful considerations to control the item difficulty consistently. There are still rooms for improvement for generative AIs to overcome vulnerabilities in developing test items. These include lack of "factual accuracy" and possible bias and sensitivity issues found in AI-generated items (Xi, 2023, p. 369). Controlling consistent item difficulty across tests is crucial as it directly impacts test reliability. Therefore, incorporating a final "human touch" is essential in the item development process to carefully control the difficulty level. Yet, more evidence that can prove whether AI-generated items edited by experienced item writers might show the same quality as those developed by human developers only is needed (Xi, 2023).

Thirdly, it would be beneficial to utilize generative AIs to develop practice materials for various other language tests. At the current stage, it is difficult to utilize the generative AI technology to create actual tests, particularly the high-stakes standardized tests. Before sufficient validation processes are carried out and a consensus is established among test developers, it would be ideal to use these AI-generated items for test preparation purposes. Students who cram for exams demand more practice items that are similar to actual tests and a great amount of time and many human resources are required to develop such practice materials. If generative AI is employed for creating test preparation materials, it could offer potential test-takers the advantage of immediate access to a substantial pool of practice items. However, careful control over content and ethical factors is required if the materials are used for commercial use.

5.3 Limitations

This study is not without limitations. The review on existing similar studies was not sufficient as this field of AI-based item generation is still in its early stage (see Attali et al., 2022). To our knowledge, there were only a handful of studies that explored the performance of generative AI such as ChatGPT for test development purpose. Only a recent study made an attempt to create English test items using ChatGPT and introduced manuals for pre-service teachers (Shin, 2023). Although few existing studies have introduced AI-generated content to examine learners' reading comprehension experiences (Lee et al., 2023) and investigated how L2 learners employ post-editing strategies with neural machine translators (Shin & Chon, 2023), there remains a gap in the research. Specifically, no study has yet attempted to create test items using generative AI while also equating those tests to traditional human-developed ones. Another limitation pertains to the selection of the instrument and participants. This study was conducted within the South Korean context, using the high-stakes university entrance exam, CSAT. Therefore, the findings might not be generalizable to other conventional language tests. Notably, a study found that ChatGPT's performance in answering the items correctly was very high in both the CSAT and the TOEFL iBT (Kwon & Lee, 2023). This implies that similar machine learning performance can also be found in item development of other conventional language tests. Lastly, the sampling of intact classes was made randomly but there was no baseline test that confirmed that they all had statistically the same proficiency. Also, we did not test and discuss the reproducibility of ChatGPT generated items, as we cannot ask the participants to take another version of B-type test. We leave it for future studies to replicate this study and delve more deeply into the reproducibility of AI-generated tests.

5.4 Suggestions for future studies

In continuation of this early study that explores the potential of using generative AI for test development, follow-up studies are crucial for further establishing these processes of developing tests. It would be beneficial to utilize generative AIs to develop tests of other language functions such as listening, speaking, or writing. Different results might be found when developing items through generative AIs to assess these language skills. Furthermore, it may be even easier and more consistent to create items with different response types, such as short answers or summaries, as these item structures are simpler. Consequently, less effort in answer option design is needed for these items, allowing very little work left for humans. Moreover, other types of generative AI technology, such as Text-To-Images, can be applied in creating items in the future. The present study only investigated the potentials of developing text passages but follow-up studies should consider creating images and videos to accompany textual passages in language tests. Most importantly, more research is required to validate the test quality and the fine-tuned control of the item difficulty of the AI-generated items.

Appendix 1

AI tools for automated item generation

Category	Tool	Website	Feature
1. AIG ^a	ChatGPT	https://chat.openai.com/	Free (Plus version is paid), creating both passages and questions at once
	Perplexity AI	https://www.perplexity.ai/	Free (GPT-4 version is paid), creating both passages and questions at once
	Poe AI (Assistant, Claude2, etc.)	https://poe.com/	Free + paid, creating both passages and questions at once, with a wide selection of LLM-based chatbots
	Q-Craft	https://q-craft.co.kr/	Paid, creating passages and questions at once, based on the standard achievement of the English National Curriculum in South Korea
	genQue	https://www.genque.ai/	Free + paid, creating passages and questions separately, with adjustable difficulty levels
2. TTI ^a	DALL.E3	https://chat.openai.com/	Paid, embedded in ChatGPT Plus, English prompt available, selecting a picture style by typing in illustration/painting/mosaic, etc
	Bing Image Creator	https://www.bing.com/create	Free, English prompt available, creation slows down when free credits are used up, but is still free (creation time is shown in seconds), using the same algorithm as DALL.E3
	MS Designer	https://designer.microsoft.com/	Free, English prompts available, creating high-quality images, easy to enter text, great for creating e-posters
	Adobe Firefly	https://www.adobe.com/kr/products/firefly.html	Free beta version, a choice of prompt languages, detailed settings for image style, tone and quality

1 Automated Item Generation

2 Text-to-Image tools

Appendix 2

The procedure for creating item generator using my GPTs.

<
KR-English Curriculum-based Item Generation
Published - g! Anyone with a link
+ Click 'Configure'

Create
Configure

Name

Description

Instructions

- If typing in an achievement standard code, create test items meeting the following conditions, please:
- Please create test items based on the Achievement Standard code, including the question + passage + options + answer choice + answer explanation + expected selection rate per option + expected rate of correct answer.
- When creating items, create three items with a different difficulty level (Low, Medium, High) at the same time.
- Use the model item types as a guide, but create an item with a new passage based on different topics.

Conversation starters

Knowledge

If you upload files under Knowledge, conversations with your GPT may include file contents. Files can be downloaded when Code Interpreter is enabled

Information for Item Gener...
PDF

Capabilities

- ☒ Web Browsing
- ☒ DALL·E Image Generation
- ☐ Code Interpreter ⓘ

				passage, make it similar to the length and the vocabulary level of the model item's passage.	
School Year	1 st Grade in High School			- Prepare a PDF document to upload to Knowledge that includes detailed additional information such as 'Test Item Card'	
Achievement Standard Information	Curriculum	☒ 2015 National Curriculum	☐ 2022 National Curriculum		
	Achievement Standard	You can read articles on general topics you're familiar with and dig into the details.			
	Achievement Standard Code	[10 영 03-01]			
Model Test Item	Learning Elements	To identify details (drawings, photos, diagrams, objects, topics)			
	Keywords	Seminar, Modern Technology, San Francisco, Keynote Speakers			
	Topic	Invitation to the annual seminar on modern technological advances			
	Item No.	E_10_3.1.1.3.1_01			
	Item Type	MCQ (Multiple-choice Question)			
	Question	What was not mentioned in the following article about seminars?			
	Passage	Dear Kelly, I am delighted to inform you that our annual seminar on modern technology progress is just around the corner. The seminar is scheduled to take place in San Francisco, California on the 15th of September. Our keynote speakers for this event are Dr. Emily Gray from Stanford University and Mr. John Lee, from a famous tech company. We expect many audiences, from students to experienced experts in the tech industry. Notably, members from top Silicon Valley firms will also be present. Please note that a detailed schedule and topics to be discussed will be sent to you a week before the event. I hope to see you there. Warm regards, James Whitmore			
	Visual aids				
	Choices	① Topic ② Place ③ Speaker ④ Food Menu ⑤ Companies attending			
	correct answer	④			
expected rate of correct answer	0%~10%	20%~30%	30%~40%		
	50%~60%	70%~80%	90%~100%		
	√				
Test Item Type: Multiple Choice				-Choose Web Browsing and DALL-E Image Generation -Choose options to publish and click 'Confirm' -By only entering a curriculum code, the	
Passage: Dear Kelly, I am delighted to inform you that our annual seminar on modern technology progress is just around the corner. The seminar is scheduled to take place in San Francisco, California, on the 15th of September. Our keynote speakers for this event are Dr. Emily Gray from Stanford University and Mr. John Lee from a famous tech company. We					

expect many audiences from students to experienced experts in the tech industry. Notably, members from top Silicon Valley firms will also be present. Please note that a detailed schedule and topics to be discussed will be sent to you a week before the event. I hope to see you there. Warm regards, James Whitmore Question: What is NOT mentioned in the above passage about the seminar? Options: 1. The topic of the seminar. 2. The location of the seminar. 3. The speakers at the seminar. 4. The food menu at the seminar. 5. The companies expected to attend. Answer: 4. The food menu at the seminar. Explanation: The passage mentions the topic, location, speakers, and expected attendees of the seminar. However, it does not mention anything about the food menu. Expected Selection Rate per Option: Option 1: 10% - 20% Option 2: 0% - 10% Option 3: 0% - 10% Option 4: 60% - 70% Option 5: 10% - 20% Expected Rate of Correct Answer: 60% - 70%	corresponding test items will be generated in high, medium, and low difficulty, respectively. -If you want to modify the generated test items, you can enter additional requirements. For example, if the passages are too short, you can also type "Make the passages a bit longer".
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Appendix 3

Questionnaire items

1. Name:
2. School Year:
3. Test Form B was AI-generated except for the last 5 questions. What were your impressions after taking the test?
- 1) I thought the items were written by some non-native speakers of English because they were very awkward
- 2) I thought the items were not written by teachers because they were somewhat awkward
- 3) They were normal
- 4) I thought the items were written by teachers because they were generally natural
- 5) I thought the items were written by professional item writers because they were very natural

4. How did you feel about the difficulty level of the items written by the AI, compared to those CSAT items?

- 1) Very easy
- 2) Somewhat easy
- 3) Similar
- 4) Somewhat difficult
- 5) Very difficult

5. If you answered 1) and 2) in Question 4, what aspects were easy for you?

- 1) Vocabulary
- 2) Sentence structure
- 3) Content
- 4) Item Complexity (answer option design)
- 5) Other

6. If you answered 4) and 5) in Question 4, what aspects were difficult for you?

- 1) Vocabulary
- 2) Sentence structure
- 3) Content
- 4) Item Complexity (answer option design)
- 5) Other

7. If AI can spontaneously generate items similar to the tests you just took, would they be helpful for your test preparation?

- 1) Not at all helpful
- 2) Not very helpful
- 3) Neutral
- 4) Somewhat helpful
- 5) Very helpful

Data availability The datasets generated and analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Competing interests The authors declare that they have no competing interests related to this work.

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